

Optimization and Numerical Methods

Weekly assignment answers

1 Solution set of Assignment:1 Topics: Convex set

1. Let x be a real-valued random variable with $\text{prob}(x = a_i) = p_i, i = 1, \dots, n$, where $a_1 < a_2 < \dots < a_n$. Now, $p \in \mathbb{R}^n$ lies in the standard probability simplex $P = \{p | 1^T p = 1, p \geq 0\}$. Which of the following conditions are convex in p ? (That is, for which of the following conditions is the set of $p \in P$ that satisfy the condition convex?).

Solution. We first note that the constraints $p_i \geq 0, i = 1, \dots, n$, define halfspaces, and $\sum_{i=1}^n p_i = 1$ defines a hyperplane, so P is a polyhedron (as P is the solution set of an inequality and equality). The first five constraints are, in fact, linear inequalities in the probabilities p_i .

- (a) $\alpha \leq Ef(x) \leq \beta$, where $Ef(x)$ is the expected value of $f(x)$, i.e.,

$$Ef(x) = \sum_{i=1}^n p_i f(a_i).$$

(The function $f : \mathbb{R} \rightarrow \mathbb{R}$ is given.) So the constraint is equivalent to two linear inequalities:

$$\alpha \leq \sum_{i=1}^n p_i f(a_i) \leq \beta.$$

implies

$$\alpha \leq \sum_{i=1}^n p_i f(a_i) \text{ and } \sum_{i=1}^n p_i f(a_i) \leq \beta$$

The set of solutions of the above system of linear inequalities corresponds to the intersection of the half-spaces defined by each of the inequalities. It is a convex set, since the half-spaces are convex sets, and the intersection of a set of convex sets is also convex. Therefore the given condition is convex in p .

- (b) $\text{prob}(x > \alpha) \leq \beta$. Now,

$$\text{prob}(x > \alpha) = \sum_{i: a_i > \alpha} p_i,$$

so the constraint is equivalent to a linear inequality:

$$\sum_{i: a_i > \alpha} p_i \leq \beta.$$

By using the similar argument as in 1.(a) on the convex half space, we can say that given condition is convex in p .

- (c) The constraint is equivalent to a linear inequality:

$$Ex^2 = \sum x_i^2 p_i \leq \alpha.$$

Use similar argument as above.

- (d) The constraint is equivalent to a linear inequality:

$$Ex^2 = \sum x_i^2 p_i \geq \alpha.$$

Use similar argument. Thus, the above four constraints, define convex sets.

- (e) $\text{var}(x) \leq \alpha$, where $\text{var}(x) = E(x - Ex)^2$ is the variance of x .

This constraint

$$\text{var}(x) = E(x - Ex)^2 = \sum a_i^2 p_i - \left(\sum p_i a_i\right)^2 \leq \alpha$$

is not convex, in general. As a counterexample, we can take $n = 2, a_1 = 0, a_2 = 1$, and $\alpha = 1/5$. The points $p = (1, 0)$ and $p = (0, 1)$ satisfy this condition, but the convex combination $p = (1/2, 1/2)$ does not satisfy, leads to $\text{var}(x) = 1/4$ (not less than $1/5$), thus doesn't satisfy the constraint.

- (f) $\text{var}(x) \geq \alpha$.

Let us consider the set $A = \{x | \text{var}(X) > \alpha\}$ which can be interpreted as the set of all random variables whose variance is greater than α . The set A would be convex if x_1 and x_2 in A , and $\theta \in (0, 1)$, implies its convex combination $x_\theta = \theta x_1 + (1 - \theta)x_2$ also lies in A . Derive the expectations $E(x_\theta)$ and $E(x_\theta)^2$. Then from those expressions derive the variance of x_θ which leads to

$$\text{var}(x_\theta) = \theta^2 \text{var}(x_1) + (1 - \theta)^2 \text{var}(x_2) + 2\theta(1 - \theta)\text{cov}(x_1, x_2),$$

Now the sign the covariance term could be negative and if it is large (negative) enough, $\text{Var}(x_\theta)$ can be smaller than both smaller than $\text{var}(x_1)$ and $\text{var}(x_2)$, violating convexity. One can formulate a counterexample where $\text{var}(x_\theta)$ go below α , implying that the set is not convex in general.

- (g) $\text{quartile}(x) \geq \alpha$, where $\text{quartile}(x) = \inf\{\beta | \text{prob}(x \leq \beta) \geq 0.25\}$.

Let us denote $\text{quartile}(x) = f(p)$ to emphasize that it is a function of p . The constraint is equivalent to

$$\text{prob}(x \leq \beta) < 0.25 \quad \text{for all } \beta < \alpha.$$

If $\alpha < a_1$, this is always true. Otherwise, define $k = \max\{i | a_i < \alpha\}$. This is a fixed integer, independent of p . The constraint holds if and only if

$$\text{prob}(x \leq a_k) = \sum_{i=1}^k p_i < 0.25.$$

This is a strict linear inequality in p , which defines an open halfspace.

- (h) $\text{quartile}(x) \leq \alpha$. The constraint is equivalent to

$$\text{prob}(x \leq \beta) \geq 0.25 \quad \text{for all } \beta \geq \alpha.$$

This can be expressed as a linear inequality:

$$\sum_{i=k+1}^n p_i \geq 0.25,$$

(If $\alpha \leq a_1$, we define $k = 0$.)

2. (a) Prove that the intersection of two convex sets is also a convex set.

Solution. Let A and B be two convex sets in a vector space V . We aim to show that their intersection $C = A \cap B$ is also a convex set.

Consider any two points $x, y \in C = A \cap B$. By the definition of set intersection, this implies that:

$$x \in A \quad \text{and} \quad x \in B,$$

$$y \in A \quad \text{and} \quad y \in B.$$

Since A is convex, for any $\lambda \in [0, 1]$, we have:

$$\lambda x + (1 - \lambda)y \in A.$$

Similarly, since B is convex, for any $\lambda \in [0, 1]$, we have:

$$\lambda x + (1 - \lambda)y \in B.$$

Therefore, for any $\lambda \in [0, 1]$, we have:

$$\lambda x + (1 - \lambda)y \in A \cap B = C.$$

Hence, C is convex.

Conclusion: The intersection of two convex sets is also a convex set.

- (b) Show that if S_1 and S_2 are two convex sets in $\mathbb{R}^{m+n}$, then so is their partial sum $S = \{(x, y_1 + y_2) : x \in \mathbb{R}^m, y_1, y_2 \in \mathbb{R}^n, (x, y_1) \in S_1, (x, y_2) \in S_2\}$.

Solution. We consider two points $(x, y_1 + y_2), (x^*, y_1^* + y_2^*) \in S$, i.e., with

$$(x, y_1) \in S_1; (x, y_2) \in S_2; (x^*, y_1^*) \in S_1; (x^*, y_2^*) \in S_2$$

For $0 \leq \theta \leq 1$,

$\theta(x, y_1 + y_2) + (1 - \theta)(x^*, y_1^* + y_2^*) = (\theta x + (1 - \theta)x^*, \theta y_1 + (1 - \theta)y_1^* + \theta y_2 + (1 - \theta)y_2^*)$ is in S because, by convexity of S_1 and S_2 ,

$$(\theta x + (1 - \theta)x^*, \theta y_1 + (1 - \theta)y_1^*) \in S_1 \quad \text{and} \quad (\theta x + (1 - \theta)x^*, \theta y_2 + (1 - \theta)y_2^*) \in S_2$$